

$$\text{Total force } F = \frac{Zmc}{hf} - \frac{Zmc}{mc^2} = \frac{Zmc}{c}$$

- 28 C** Since $\Delta E = hf$, from energy level diagram, the energy difference is lesser towards bigger f (emission from higher levels).

Wavelength is inversely proportional to frequency, hence the spectrum is in opposite order.

- 29 C** Let C be original number of counts due to the α -source.

Let B be the background count.

$$C + B = 300 \quad \text{----- (1)}$$

$$0.5C + B = 155 \quad \text{----- (2)}$$

Solving, $C = 290$

Lead would block all the counts due to the α -source.

Hence, only background count of 10 is detected.

- 30 B**